

TEJ GUMASTE

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EDUCATION

University of Kansas

Bachelor of Science in Computer Science and Minor in Mathematics - GPA 3.85

Lawrence, KS

Expected graduation May 2026

Awards

SELF fellow, University Scholar (Top 5%), IEA Full Scholarship Recipient (Top 2%), Babcock Math Scholarship (Top 5%)

TECHNICAL SKILLS

Languages: Java, Python, C#, C++, C, MatLab, Solidity, Javascript, Typescript, Scala

Technologies: Git, VS Code, Visual Studio, Android Studio, PyCharm, Docker, Unity, PostgreSQL, Playwright, Forge

Frameworks: React, NextJS, Svelte, Agile, ExpressJS

Machine Learning: Logistic Regression, Feature Selection (RFE), TF-IDF, Model Evaluation

Data Processing: NumPy, Pandas, Matplotlib, Sklearn

EXPERIENCE

Software Engineering Intern - Funding & Payments

Jun. 2025 - Aug. 2025

Gemini Exchange

New York City, NY

- Co-developed the backend integration of CrossRiver Bank wire payment rails, handling webhook events with **AWS SQS** using an MVC architecture in **Scala**, and validation flows—supporting an **expected \$4M** in additional annual revenue
- Streamlined Gemini's tax reconciliation by enhancing services to merge **10,000+ daily crypto rates**, achieving **100% coverage**, eliminating **4+** annual tax reprocesses and all manual hot fixes
- Engineered **5+** smoke tests using **Playwright** for web and mobile platforms that reduced deployment lifecycle by **15%** and polished user-facing web and mobile views using **Typescript**

Research Engineer - First Author, Peer Reviewed

Feb. 2024 – Present

Blakemore Lab - KU

Lawrence, KS

- Developed data engineering pipelines to efficiently process and visualize atomic properties of **1500+** structures using novel parsing techniques, published as **first author in Molecules Journal**
- Detected the presence and effects of metals within the crystallographic files through large scale data analysis using **NumPy**
- Automated the analysis workflow for crystallographic structures, reducing the time from **6 months to 5 minutes**

Software Engineering Intern

Jun. 2023 – Dec. 2023

Security Benefit

Topeka, KS

- Automated Sailpoint access management by building custom APIs, resulting in a time savings of **100 hours per year**
- Developed and maintained ETL processes using Python and **AWS (S3)** to support internal and consumer applications

PROJECTS

TapGuard - View Project | NextJS, NFC, Android, ExpressJS, Solidity, Forge, TRON

Feb. 2026

- Engineered a payment ecosystem utilizing ExpressJS, Socket.io and NFC to achieve seamless "Tap to Pay" crypto checkouts for retail environments.
- Architected a high-performance dual-chain settlement engine to process stablecoin payments with automated backend reconciliation on the TRON network, significantly reducing transaction overhead and latency for merchants.

LookSmith - View Project | NextJS, Tailwind CSS, PostgreSQL, Godot, OpenCV/MediaPipe, Websockets

Oct. 2025

- Engineered a MediaPipe/OpenCV (Python) facial tracking pipeline that recognizes **3+ facial gestures** with **95%** accuracy
- Developed a **Socket.io** connection and a data transfer protocol to translate openCV pipeline data into 3D editor actions

EirAI - HackKU25 Theme Track 2nd Place - View Project | NextJS, PostgreSQL, OpenAI, Gemini

Apr. 2025

- Co-developed EirAI, a tool that transcribes patient-doctor conversations into SOAP notes, reducing administrative bloat
- Implemented semantic indexing for **73K ICD-10 codes**, matched to treatments using a simulated dataset (**1.5M+ rows**)
- Engineered approval prediction module using historical provider data to estimate insurance claim success rates

DAIM - AI Track Winner - View Project | NextJS, Tailwind CSS, Pinata IPFS, Metamask, Autonoms Auto Drive

Mar. 2025

- Developed a full-stack Web3 platform enabling secure, open-source sharing of AI models using blockchain signing and IPFS
- Integrated generative AI tools for code summarization and model context to enhance user experience and discovery

LEADERSHIP

VP Engineering and Education

Aug. 2023 – Apr. 2024

KU Blockchain Institute (KUBI)

Lawrence, KS

- Spearheaded bi-weekly blockchain (Ethereum) and web development (React, NextJS, Express) workshops for groups of **15+** students, utilizing hands-on projects and real-world examples to enhance learning outcomes
- Engineered smart contracts for KUBI's Decentralized Autonomous Organization that logged over **500 hours** of club participation